MACHINE LEARNING ITA0609
SDG ASSIGNMENT

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Design of a Machine Learning System to Reduce Industrial Waste

SDG 12 - Responsible Consumption and Production

Assignment Question 28

Assignment Question	Design a Machine Learning system to reduce industrial waste.

1. Problem Understanding

Problem Statement: Industrial production produces scrap material, rejected products, excess packaging, chemical residue, unused raw material and energy loss. In many factories, waste records are reviewed only after the production batch is completed. This delays corrective action and increases production cost, disposal requirements and environmental pollution. The proposed system uses Machine Learning to study production conditions, estimate waste risk and identify the factors that are most likely to cause excessive waste.

The objective is to develop a data-driven decision-support system that predicts waste quantity or waste level before a batch is completed, provides early alerts and recommends suitable actions to reduce material loss.

The Machine Learning system does not physically remove industrial waste. It analyses data obtained from machine sensors, quality-inspection records, inventory systems and production reports, and helps the production team take preventive action.

Objectives:

- Collect production, machine, quality and waste-related data.
- Clean and preprocess the collected industrial data.
- Perform exploratory data analysis to understand waste patterns.
- Predict waste quantity and classify waste risk as Low, Medium or High.
- Identify important factors responsible for defects and scrap.
- Generate alerts and recommendations for production operators.
- Support SDG 12 through efficient use of materials and responsible production.

2. SDG Mapping

Sustainable Development Goal: SDG 12 - Responsible Consumption and Production.

Connection:

- Improving raw-material efficiency and reducing unnecessary consumption.
- Preventing avoidable defects and rejected products.
- Supporting reuse, recycling and correct waste segregation.
- Reducing disposal cost, pollution and environmental impact.
- Providing measurable information for sustainable production planning.

3. Dataset

Recommended Dataset: A manufacturing or industrial waste dataset collected from a factory, Kaggle, UCI Machine Learning Repository, or an organisation's production database. For academic demonstration, a synthetic dataset can also be created using realistic machine and production ranges.

Typical Attributes:

Attribute	Meaning
Batch_ID	Unique production-batch number

Machine_Type	Machine or production-line category
Raw_Material_kg	Raw material used for the batch
Temperature_C	Machine operating temperature
Pressure_bar	Production pressure
Speed_rpm	Machine operating speed
Material_Quality	Quality score of incoming material
Energy_kWh	Energy consumed during production
Downtime_min	Machine downtime
Defect_Rate	Percentage of rejected or defective items
Waste_kg	Waste generated per batch
Waste_Risk	Low, Medium or High waste-risk class

Target Variable: Waste_kg for regression, or Waste_Risk / Waste_Category for classification.

4. Data Collection and Preparation

Workflow:

1. Import NumPy, Pandas and the required Machine Learning libraries.
2. Load production data from CSV files, sensors, quality reports or databases.
3. Display the first records and verify dataset shape and column names.
4. Check for missing, duplicate and invalid values.
5. Replace or remove missing records using suitable methods.
6. Encode categorical columns such as machine type and waste category.
7. Create derived features such as waste percentage, defect rate and material efficiency.
8. Select the important features required for model training.

5. Exploratory Data Analysis

The following analyses help in understanding the industrial process before model training:

- Dataset shape, column information and data types.
- Statistical summary using mean, minimum, maximum and standard deviation.
- Waste-risk class counts and waste-category frequencies.
- Average waste by machine type, shift or product category.
- Relationship between temperature, pressure, speed, defect rate and waste quantity.
- Correlation analysis to identify strongly related numerical features.

6. Visualization and Interpretation

Useful visualizations include waste distribution, average waste by machine type, defect trends, correlation heatmap and feature-importance graph. The following charts use an illustrative synthetic dataset and are included only to demonstrate the expected analysis format.

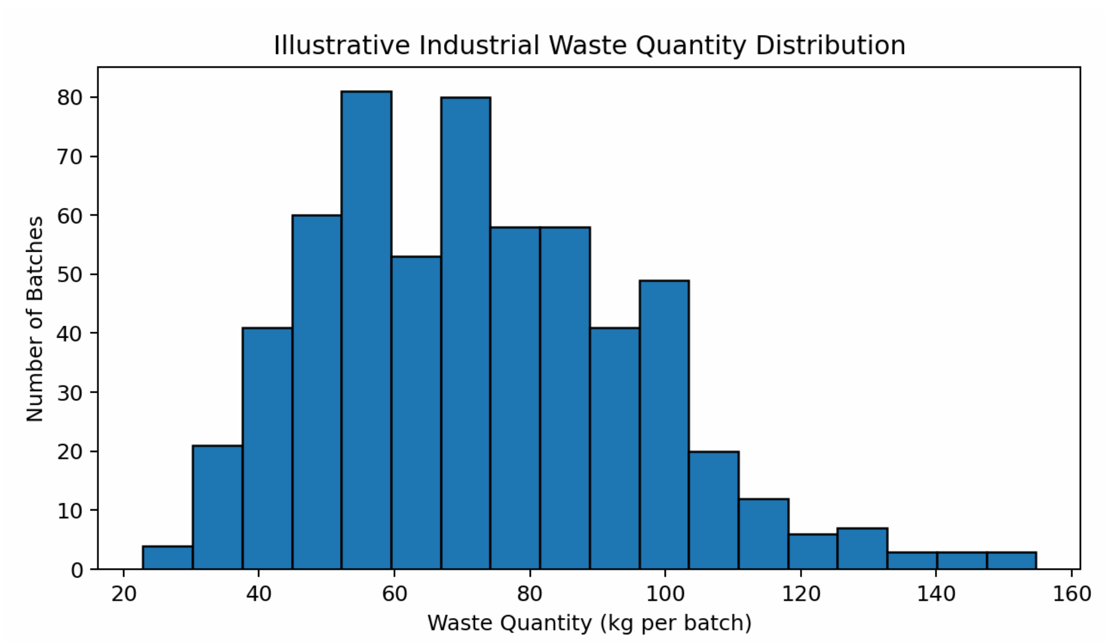

Figure 1. Illustrative distribution of industrial waste quantity per batch

Interpretation: Most production batches generate a moderate amount of waste, while a smaller number of batches show unusually high waste. These high-waste batches require detailed investigation and preventive action.

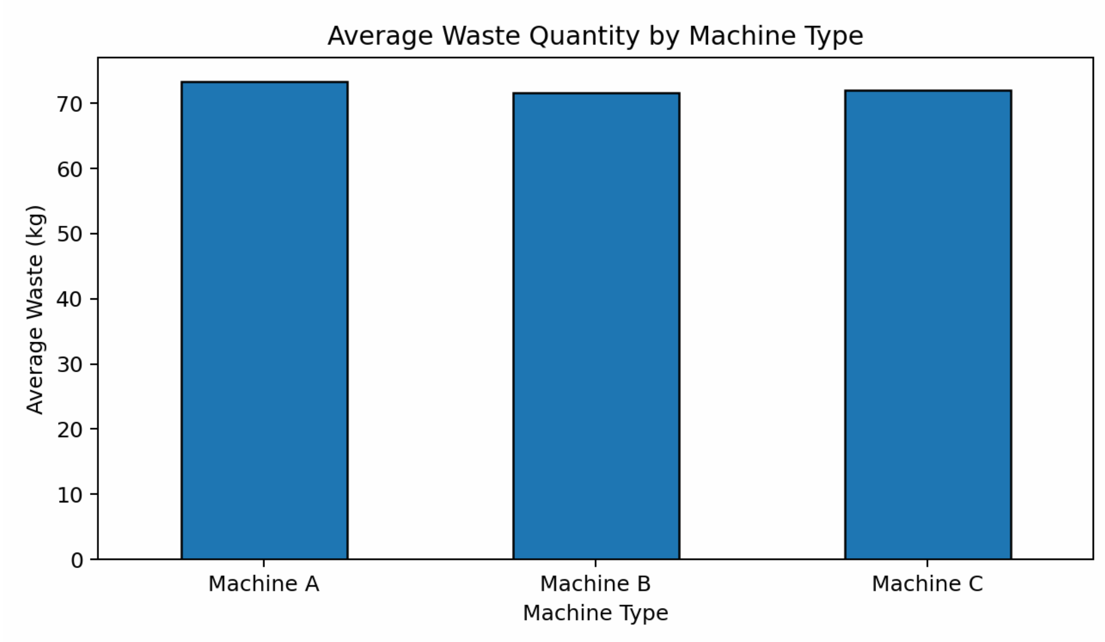

Figure 2. Illustrative comparison of average waste across machine types

Interpretation: Comparing machines helps identify the production line that requires maintenance, calibration or process improvement.

7. Proposed Machine Learning System

The proposed system combines production-data collection, preprocessing, Machine Learning prediction, alerts and reporting. It can be implemented as a desktop application, web dashboard or cloud-based monitoring system.

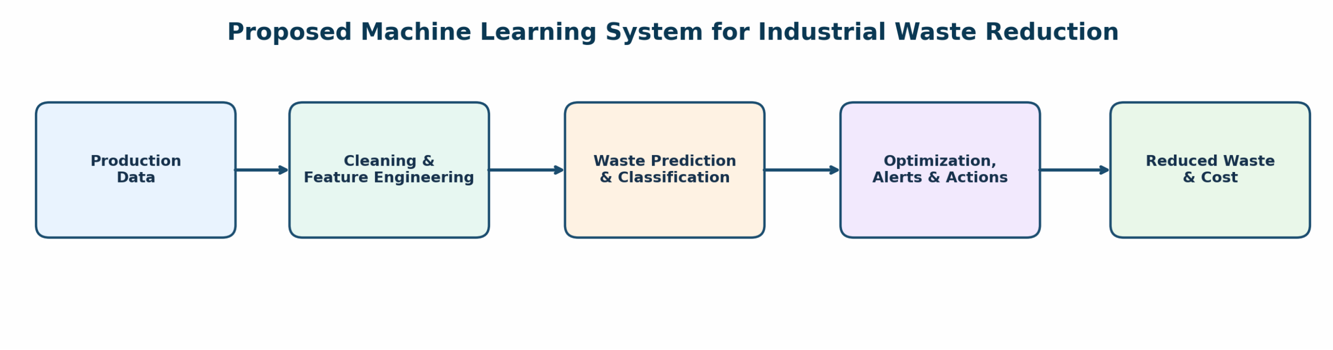

Figure 3. Workflow of the proposed Machine Learning system for industrial waste reduction

8. Three Main Modules

Module 1: Data Collection and Preprocessing

This module collects values from sensors, machine logs, inventory records and quality-inspection reports. It removes duplicate or incorrect records, handles missing values, encodes categorical features and calculates useful indicators such as defect rate, waste ratio and material efficiency.

Module 2: Waste Prediction and Classification

This module uses cleaned data to train the Machine Learning model. A regression model can estimate the expected waste quantity in kilograms, while a classification model can predict Low, Medium or High waste risk. The model can also classify the likely waste type, such as material scrap, defective product, packaging waste or chemical residue.

Module 3: Optimization, Alert and Reporting

This module compares the predicted result with an acceptable waste limit. When excessive waste is expected, the system displays an alert and highlights the influential parameters. It can recommend actions such as adjusting temperature or speed, checking raw-material quality, inspecting the machine or scheduling maintenance. Results are shown through a dashboard and can be downloaded as reports.

9. Machine Learning Model

Recommended Algorithm: Random Forest Classifier / Random Forest Regressor.

Reasons:

- Works well with industrial tabular data.
- Handles nonlinear relationships among machine parameters.
- Can perform both classification and regression.
- Reduces overfitting compared with a single decision tree.
- Provides feature-importance scores for interpretation.

- Handles multiple features without requiring a complex mathematical model.

Alternative Algorithms:

- Decision Tree
- Gradient Boosting
- Support Vector Machine
- Linear or Logistic Regression
- Artificial Neural Network

10. Train-Test Split and Model Evaluation

The dataset is commonly divided into 80% training data and 20% testing data. The model learns production patterns from the training set and is evaluated using unseen testing records. Cross-validation may be applied to compare parameter settings and improve reliability.

Task	Evaluation Measures
Waste-risk classification	Accuracy, Precision, Recall, F1 Score, Confusion Matrix and Classification Report
Waste-quantity prediction	Mean Absolute Error, Mean Squared Error, Root Mean Squared Error and R-squared
Operational impact	Reduction in scrap, rejected products and rework percentage
Business impact	Material cost saved, energy saved and disposal cost reduced

11. Illustrative Python Implementation

The following simplified code shows how a Random Forest classifier can be trained to predict the waste-risk class. The exact columns should be changed according to the selected dataset.

```
import pandas as pd
from sklearn.model_selection import train_test_split
from sklearn.ensemble import RandomForestClassifier
from sklearn.metrics import accuracy_score, classification_report

data = pd.read_csv("industrial_waste.csv")

X = data[["Temperature_C", "Pressure_bar", "Speed_rpm",
          "Material_Quality", "Energy_kWh", "Downtime_min",
          "Defect_Rate"]]
y = data["Waste_Risk"]

X_train, X_test, y_train, y_test = train_test_split(
    X, y, test_size=0.20, random_state=42, stratify=y
)

model = RandomForestClassifier(n_estimators=200, random_state=42)
model.fit(X_train, y_train)

prediction = model.predict(X_test)
print("Accuracy:", accuracy_score(y_test, prediction))
print(classification_report(y_test, prediction))
```

Note: The implementation results depend on the selected dataset, its size, data quality, class balance and feature engineering. The charts and metrics below are illustrative and should not be treated as certified factory performance.

12. Example Dataset Records

Machine	Material kg	Temp °C	Pressure	Speed	Defect Rate	Waste kg	Risk
Machine A	622.2	105.2	6.1	895.0	0.233	43.7	Medium
Machine C	1467.3	94.9	6.6	1381.0	0.172	97.2	Low
Machine A	654.1	133.5	9.1	667.0	0.305	62.5	High
Machine A	1391.0	94.3	2.7	1524.0	0.341	138.1	High
Machine A	801.3	76.6	7.4	1146.0	0.121	51.6	Low
Machine A	1492.8	65.0	4.2	551.0	0.37	140.6	High

13. Illustrative Model Results

Metric	Illustrative Result
Accuracy	80.0%
Weighted Precision	80.6%
Weighted Recall	80.0%
Weighted F1 Score	78.7%

These results were generated from a synthetic demonstration dataset. A real industrial implementation must be trained and validated using factory-specific production records.

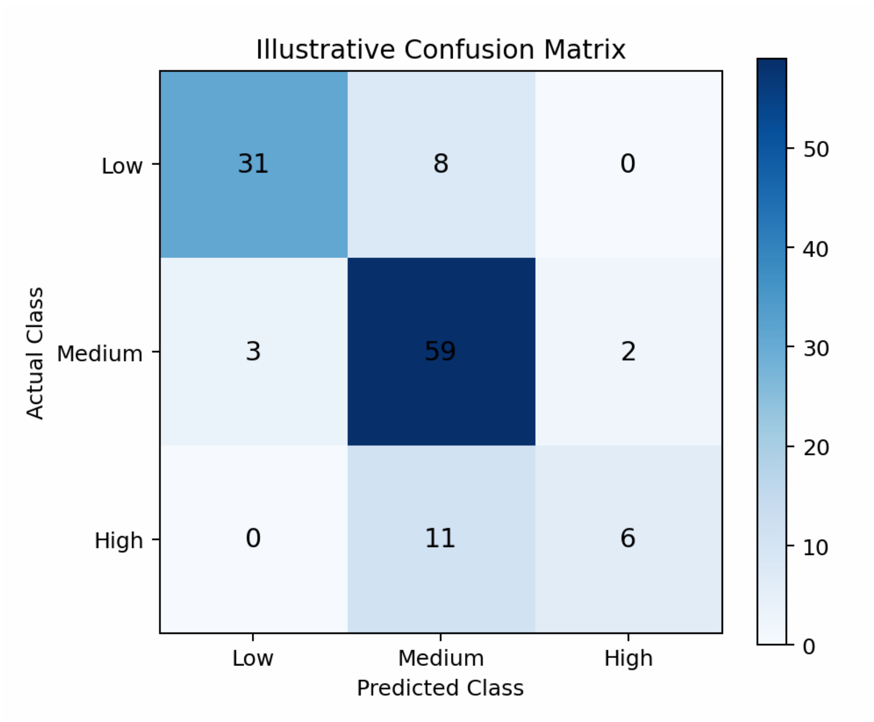

Figure 4. Illustrative confusion matrix for Low, Medium and High waste-risk classes

14. Feature Importance

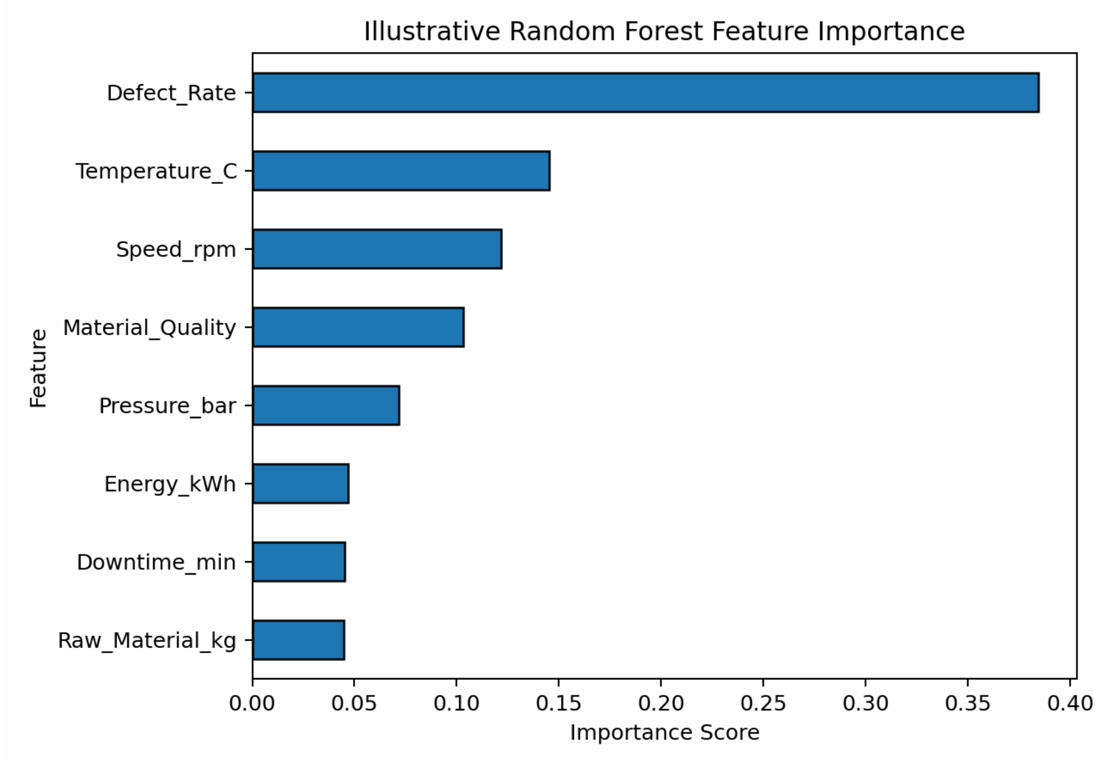

Figure 5. Illustrative Random Forest feature-importance graph

Interpretation: Features with higher importance have a stronger influence on model decisions. This information helps production engineers identify which machine or material conditions should be controlled first.

15. Expected Output

The proposed dashboard should display:

- Predicted waste quantity for the current production batch.
- Waste-risk level: Low, Medium or High.
- Likely cause or waste category.
- Important parameters contributing to the prediction.
- Alert message when the predicted waste exceeds the acceptable limit.
- Recommended corrective action.
- Graphs for waste trends, machine comparison and feature importance.
- Downloadable production and prediction report.

16. Example Working Scenario

Consider a plastic manufacturing unit. The system receives raw-material quantity, machine temperature, mould pressure, speed, energy consumption, downtime and defect rate. The model predicts a High waste risk because temperature and speed are outside the normal range and material quality is low. The dashboard immediately displays an alert and recommends reducing the speed, checking mould alignment and verifying raw-material quality. The operator can act before the complete batch becomes defective.

17. Innovation

Innovative Features:

- Waste prediction before completion of the production batch.
- Automatic detection of abnormal machine conditions.
- Feature-importance-based explanation of the prediction.
- Waste segregation and recycling recommendation.
- Real-time dashboard alerts using IoT sensor data.
- Future waste trend and production-efficiency forecasting.
- Integration with maintenance and inventory systems.

18. SDG 12 Impact

The proposed system contributes to SDG 12 by:

- Reducing unnecessary raw-material consumption.
- Preventing defects, scrap and avoidable rework.
- Improving resource, energy and production efficiency.
- Supporting reuse, recycling and correct waste handling.
- Reducing pollution and waste sent to disposal facilities.
- Providing data for responsible production and sustainability reporting.

19. Limitations and Future Scope

The accuracy of the model depends on the quality and quantity of industrial data. Sensor faults, missing records, unbalanced waste classes and changes in production conditions can reduce prediction accuracy. The model should be retrained periodically. Future work can include real-time IoT integration, computer-vision-based defect detection, cloud dashboards, automatic machine control, carbon-emission estimation and integration with Enterprise Resource Planning systems.

20. Conclusion

The proposed Machine Learning system provides a practical method for reducing industrial waste. It collects and preprocesses production data, predicts waste quantity or risk, identifies influential factors and produces alerts and recommendations. Early prediction enables the production team to reduce scrap, lower cost, improve product quality and use resources responsibly. Therefore, the system directly supports SDG 12 - Responsible Consumption and Production.

21. Recommended Folder Structure

```
Industrial Waste Reduction System
│
├── Industrial_Waste_Prediction.ipynb
├── industrial_waste.csv
├── app.py
├── trained_model.pkl
├── README.md
├── graphs
│   ├── waste_distribution.png
│   ├── feature_importance.png
│   └── confusion_matrix.png
└── report.docx
```

22. Rubric-wise Coverage

Rubric Area	Coverage in the Assignment
Problem Understanding	Clear problem statement, objectives, need and system limitations.
Data Collection & Preparation	Industrial data sources, missing-value handling, encoding and feature engineering.
Analysis using NumPy/Pandas	Dataset shape, information, summary, value counts and correlation.
Visualization & Interpretation	Waste distribution, machine comparison, confusion matrix and feature importance.
Machine Learning	Random Forest model, train-test split and evaluation metrics.
Innovation & SDG Mapping	Early alerts, recommendations, recycling support and SDG 12 linkage.

23. Suggested References

- United Nations Sustainable Development Goal 12 - Responsible Consumption and Production.
- Scikit-learn documentation for Random Forest classification and regression.
- Pandas and NumPy documentation for data cleaning and analysis.
- Manufacturing, predictive-maintenance and industrial-waste datasets from Kaggle or UCI repositories.
- Factory sensor logs, quality-inspection reports and inventory records for real implementation.

Final Note: The proposed model is an academic design. A real industrial deployment must use factory-specific data, validate predictions with production engineers and follow the organisation's safety, quality and environmental policies.